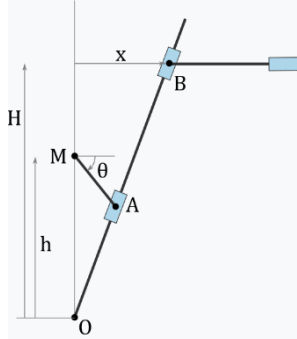


Problem 1

Problem 1

A quick-return mechanism, or Whitworth mechanism, is a mechanism that uses clever kinematics to speed up the cycle of a machine only on the return. This was developed during the industrial revolution, and helped reduce the time requirements of a wide variety of machines. The collars A and B slide freely along the bar OAB (these could also be pins in a slot), and a rotary input (motor) is included at the point M. To visualize its motion, take a look at the following GIF: [Quick Return Mechanism](#)



- Write an equation of constraint, ϕ_1 for the Whitworth machine, using $q_1 = x$ and $q_2 = \theta$ as the generalized coordinates. Note that this is a redundant set because it's only a one degree-of-freedom system.
- Take a time derivative of your equation of constraint $\dot{\phi}_1$ and find the coefficients a_{11} and a_{12} .
- Rearrange $\dot{\phi}_1$ to find an expression for $\dot{x}$ as a function of $x, \theta, \dot{\theta}$.
- Over a range of $\theta \in [0, 2\pi]$, plot x vs $\dot{x}$ assuming that $\dot{\theta}$ is constant. You should show that when $\dot{x}$ is positive, motion is slow and approximately constant, but when $\dot{x}$ is negative, it moves much faster.

Setup

Let r be the crank length MA , h the height of M above O , and H the height of the slot above O . With $q_1 = x$, $q_2 = \theta$,

$$\vec{r}_{OA} = r \cos \theta \hat{n}_1 + (h - r \sin \theta) \hat{n}_2, \quad \vec{r}_{OB} = x \hat{n}_1 + H \hat{n}_2$$

Part (a): Constraint Equation

Since O, A, B are collinear, $\vec{r}_{OA} \times \vec{r}_{OB} = \vec{0}$:

$$r \cos \theta \cdot H - (h - r \sin \theta)x = 0$$

$$\phi_1 : \quad x(h - r \sin \theta) - Hr \cos \theta = 0$$

Part (b): Coefficients a_{11} , a_{12}

$$\begin{aligned} \dot{\phi}_1 &= \frac{\partial \phi_1}{\partial x} \dot{x} + \frac{\partial \phi_1}{\partial \theta} \dot{\theta} = a_{11} \dot{x} + a_{12} \dot{\theta} \\ \frac{\partial \phi_1}{\partial x} &= h - r \sin \theta, \quad \frac{\partial \phi_1}{\partial \theta} = r(H \sin \theta - x \cos \theta) \end{aligned}$$

$$a_{11} = h - r \sin \theta, \quad a_{12} = r(H \sin \theta - x \cos \theta)$$

Part (c): Solving for $\dot{x}$

$$\dot{\phi}_1 = 0 \Rightarrow \dot{x} = -\frac{a_{12}}{a_{11}}\dot{\theta}$$

Substituting $x = \frac{Hr \cos \theta}{h - r \sin \theta}$ from part (a) into a_{12} :

$$a_{12} = r(H \sin \theta - x \cos \theta) = \frac{rH(h \sin \theta - r)}{h - r \sin \theta}$$

so that

$$\dot{x} = -\frac{a_{12}}{a_{11}}\dot{\theta} = -\frac{rH(h \sin \theta - r)}{(h - r \sin \theta)^2}\dot{\theta}$$

$$\dot{x} = \frac{Hr(r - h \sin \theta)}{(h - r \sin \theta)^2}\dot{\theta}$$

Part (d): Numerical Results

$\dot{\theta} = 5 \text{ rad/s}$ (constant), $r = 1 \text{ m}$, $h = 10 \text{ m}$, $H = 20 \text{ m}$, $\theta \in [0, 2\pi]$.

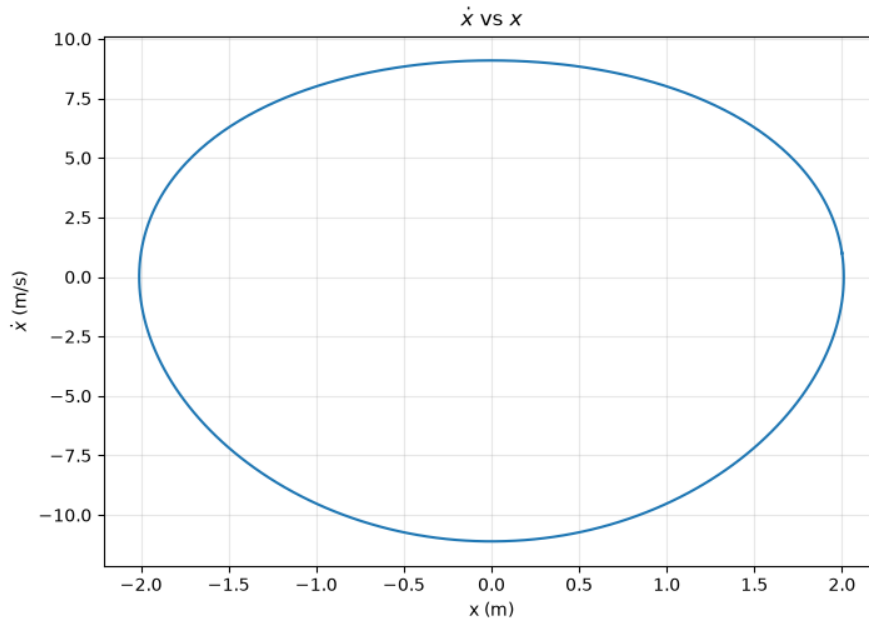


Figure 1: $\dot{x}$ vs. x over one full crank revolution.

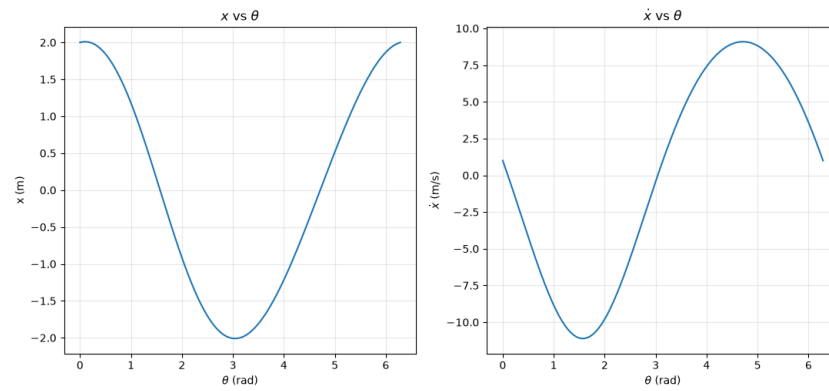


Figure 2: x and $\dot{x}$ vs. θ . Near $\theta = \pi/2$, $\dot{x}$ reaches its most negative value (brief, fast return); near $\theta = 3\pi/2$, $\dot{x}$ reaches a smaller-magnitude positive peak spread over more of the cycle (slow stroke) — the quick-return behavior of the Whitworth mechanism.

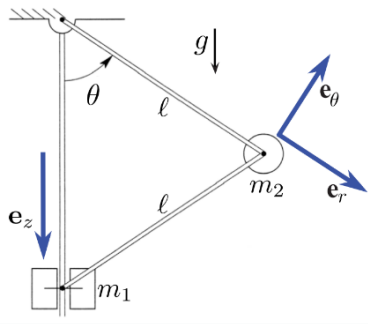
Problem 2

Problem 2

Two masses are constrained by the three bar linkages shown. Mass m_1 is constrained to move in the vertical direction while mass m_2 moves in a circular motion. Use θ as the generalized coordinate.

1. Write the position of each particle, $\vec{r}_1$ and $\vec{r}_2$, with respect to the pivot on the ceiling.
2. Calculate the inertial accelerations of each particle, $\ddot{\vec{r}}_1$ and $\ddot{\vec{r}}_2$.
3. Calculate the virtual displacement vector of each particle, $\delta\vec{r}_1$ and $\delta\vec{r}_2$, in terms of the generalized coordinate θ and a virtual displacement in θ , $\delta\theta$.
4. Write the applied force on each particle, $\vec{F}_{A1}$ and $\vec{F}_{A2}$.
5. Write the differential equation of motion for the system using D'Alembert's principle, and collect all terms.
6. This device performs oscillations when released from rest at small θ angles. What is the frequency of the oscillations in Hz? Express your answer in terms of g , ℓ , m_1 , and m_2 .

HINT: If θ is small, then $\sin\theta \approx \theta$ and $\cos\theta \approx 1$. Additionally, higher order terms such as θ^2 and $\dot{\theta}^2\theta$ are approximately zero. Finally, to find the natural frequency, get a differential equation in the form of $\ddot{x} + \omega_n^2 x = 0$. Taking the square-root of the position coefficient will give you your natural frequency.



Positions

With $\hat{e}_z$ pointing down, $\hat{e}_x$ horizontal, and θ measured from $\hat{e}_z$ so that $\hat{e}_r = \sin\theta \hat{e}_x + \cos\theta \hat{e}_z$,

$$\vec{r}_1 = 2\ell \cos\theta \hat{e}_z, \quad \vec{r}_2 = \ell \hat{e}_r = \ell \sin\theta \hat{e}_x + \ell \cos\theta \hat{e}_z$$

Velocities and Inertial Accelerations

$$\dot{\vec{r}}_1 = -2\ell \sin\theta \dot{\theta} \hat{e}_z, \quad \dot{\vec{r}}_2 = \ell \cos\theta \dot{\theta} \hat{e}_x - \ell \sin\theta \dot{\theta} \hat{e}_z$$

$$\ddot{\vec{r}}_1 = -2\ell(\dot{\theta}^2 \cos\theta + \ddot{\theta} \sin\theta) \hat{e}_z$$

$$\ddot{\vec{r}}_2 = \ell(\ddot{\theta} \cos\theta - \dot{\theta}^2 \sin\theta) \hat{e}_x - \ell(\ddot{\theta} \sin\theta + \dot{\theta}^2 \cos\theta) \hat{e}_z$$

Virtual Displacements

$$\delta\vec{r}_i = \frac{\partial\vec{r}_i}{\partial\theta} \delta\theta$$

$$\delta\vec{r}_1 = -2\ell \sin\theta \delta\theta \hat{e}_z, \quad \delta\vec{r}_2 = \ell \cos\theta \delta\theta \hat{e}_x - \ell \sin\theta \delta\theta \hat{e}_z$$

Applied Forces

$$\vec{F}_{A1} = m_1 g \hat{e}_z, \quad \vec{F}_{A2} = m_2 g \hat{e}_z$$

Equation of Motion (D'Alembert's Principle)

$$\sum_{i=1}^2 (\vec{F}_{Ai} - m_i \ddot{\vec{r}}_i) \cdot \gamma_{i1} = 0, \quad \gamma_{i1} = \frac{\partial \dot{\vec{r}}_i}{\partial \dot{\theta}} = \frac{\partial \vec{r}_i}{\partial \theta}$$

$$\gamma_{11} = -2\ell \sin \theta \hat{e}_z, \quad \gamma_{21} = \ell \cos \theta \hat{e}_x - \ell \sin \theta \hat{e}_z$$

Projecting each particle's contribution onto γ_{i1} :

$$(\vec{F}_{A1} - m_1 \ddot{\vec{r}}_1) \cdot \gamma_{11} = -2\ell m_1 g \sin \theta - 4\ell^2 m_1 \sin \theta (\dot{\theta}^2 \cos \theta + \ddot{\theta} \sin \theta)$$

$$(\vec{F}_{A2} - m_2 \ddot{\vec{r}}_2) \cdot \gamma_{21} = -\ell m_2 g \sin \theta - \ell^2 m_2 \ddot{\theta}$$

Adding, dividing by $-\ell$, and collecting terms in $\ddot{\theta}$, $\dot{\theta}^2$, and $\sin \theta$:

$$(m_2 + 4m_1 \sin^2 \theta) \ddot{\theta} + (4m_1 \ell \sin \theta \cos \theta) \dot{\theta}^2 + (2m_1 + m_2) g \sin \theta = 0$$

Natural Frequency

For small θ : $\sin \theta \approx \theta$, $\cos \theta \approx 1$, and $\sin^2 \theta$, $\dot{\theta}^2 \theta$ terms are negligible, so

$$m_2 \ell \ddot{\theta} + (2m_1 + m_2) g \theta = 0 \Rightarrow \omega_n^2 = \frac{(2m_1 + m_2) g}{m_2 \ell}$$

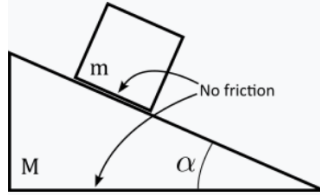
$$f = \frac{1}{2\pi} \sqrt{\frac{(2m_1 + m_2) g}{m_2 \ell}}$$

Problem 3

Problem 3

The block, with mass m , begins to slide down the ramp, with mass M , at an angle α . Use D'Alembert's principle to calculate the equations of motion in terms of the generalized coordinates: the distance of the block down the ramp x_B and the distance of the ramp from where it begins x_R .

Note that this was the example problem from our lecture on Newtonian mechanics, but we don't have to calculate constraint forces when we approach it this way.



Positions

With generalized coordinates x_R (ramp) and x_B (block, measured down the ramp),

$$\vec{r}_1 = x_R \hat{n}_1, \quad \vec{r}_2 = (x_R + x_B \cos \alpha) \hat{n}_1 + (y_0 - x_B \sin \alpha) \hat{n}_2$$

Velocities and Accelerations

$$\dot{\vec{r}}_1 = \dot{x}_R \hat{n}_1, \quad \dot{\vec{r}}_2 = (\dot{x}_R + \dot{x}_B \cos \alpha) \hat{n}_1 - \dot{x}_B \sin \alpha \hat{n}_2$$

$$\ddot{\vec{r}}_1 = \ddot{x}_R \hat{n}_1, \quad \ddot{\vec{r}}_2 = (\ddot{x}_R + \ddot{x}_B \cos \alpha) \hat{n}_1 - \ddot{x}_B \sin \alpha \hat{n}_2$$

Virtual-Displacement Coefficients

$$\gamma_{ij} = \frac{\partial \dot{\vec{r}}_i}{\partial \dot{q}_j}$$

$$\gamma_{11} = \hat{n}_1, \quad \gamma_{12} = \vec{0}, \quad \gamma_{21} = \hat{n}_1, \quad \gamma_{22} = \cos \alpha \hat{n}_1 - \sin \alpha \hat{n}_2$$

Applied Forces

$$\vec{F}_{A1} = -Mg \hat{n}_2, \quad \vec{F}_{A2} = -mg \hat{n}_2$$

Equations of Motion (D'Alembert's Principle)

For $q_1 = x_R$: since $\vec{F}_{A1}, \vec{F}_{A2} \perp \hat{n}_1$, gravity drops out and

$$(-M\ddot{\vec{r}}_1) \cdot \hat{n}_1 + (-m\ddot{\vec{r}}_2) \cdot \hat{n}_1 = 0$$

$$(M + m)\ddot{x}_R + m \cos \alpha \ddot{x}_B = 0$$

For $q_2 = x_B$: only particle 2 contributes, since $\gamma_{12} = \vec{0}$,

$$(\vec{F}_{A2} - m\ddot{\vec{r}}_2) \cdot \gamma_{22} = 0 \Rightarrow mg \sin \alpha - m(\ddot{x}_R \cos \alpha + \ddot{x}_B) = 0$$

$$\ddot{x}_B + \ddot{x}_R \cos \alpha = g \sin \alpha$$

Solving the System

Eliminating $\ddot{x}_R$ between the two boxed equations above:

$$\ddot{x}_R = -\frac{mg \sin \alpha \cos \alpha}{M + m \sin^2 \alpha}$$

$$\ddot{x}_B = \frac{(M + m)g \sin \alpha}{M + m \sin^2 \alpha}$$

Appendix: Code Listings

Problem 1 Code

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 theta_dot = 5 # theta_dot (rad/s), assumed constant
5 r = 1 # crank length MA (m)
6 h = 10 # height of M above O (m)
7 H = 20 # height of slot (B) above O (m)
8
9 theta_span = (0, 2 * np.pi)
10 theta_eval = np.linspace(theta_span[0], theta_span[-1], 500)
11
12 x_points = (H * r * np.cos(theta_eval)) / (h - r * np.sin(theta_eval))
13 x_dot_points = (H * r * (r - h * np.sin(theta_eval)) * theta_dot) / (h - r * np.sin(theta_eval)) **
14     2
15
16 fig1, ax_main = plt.subplots(figsize=(8, 5.5))
17
18 ax_main.plot(x_points, x_dot_points)
19
20 ax_main.set_xlabel('x (m)')
21 ax_main.set_ylabel(r'$\dot{x}$ (m/s)')
22 ax_main.set_title(r'$\dot{x}$ vs $x$')
23 ax_main.grid(True, alpha=0.3)
24
25 fig2, (ax_x, ax_xdot) = plt.subplots(1, 2, figsize=(13, 5.5))
26
27 ax_x.plot(theta_eval, x_points)
28 ax_x.set_xlabel(r'$\theta$ (rad)')
29 ax_x.set_ylabel('x (m)')
30 ax_x.set_title(r'$x$ vs $\theta$')
31 ax_x.grid(True, alpha=0.3)
32
33 ax_xdot.plot(theta_eval, x_dot_points)
34 ax_xdot.set_xlabel(r'$\theta$ (rad)')
35 ax_xdot.set_ylabel(r'$\dot{x}$ (m/s)')
36 ax_xdot.set_title(r'$\dot{x}$ vs $\theta$')
37 ax_xdot.grid(True, alpha=0.3)
38
39 plt.show()

```

Listing 1: Numerical evaluation of x and $\dot{x}$ vs. θ for the Whitworth quick-return mechanism, and generation of Figures 1–2.